and light, so it produces a 'bright' sound. It has good damping property, so when music stops the driver also stops. As it is light, it moves quickly and easily. Being light and rigid, aluminium dome can capture subtleness of music, like delicate brush strokes across the surface of cymbals, with amazing accuracy. Its crowned voice coil bobbins and silver-plated pole piece can extend the bandwidth over an octave above the limits of human hearing.

The arrangement of housing of tweeter and mid-range speakers is shown in Fig. 9. The small frontal area of the mid-range speaker maintains a wide dispersion throughout the unit's operating range, and the effects of the

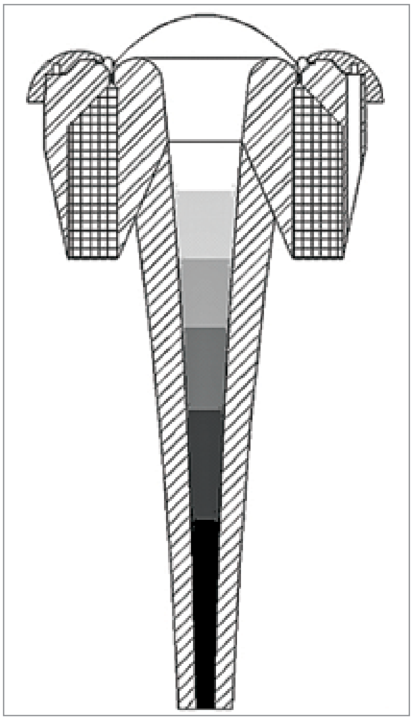

Fig. 9: Housing of tapered tube tweeter and mid-range drivers

diffraction are reduced by chamfering the edges. The design of the tweeter enclosures is also based on the fact, that if it is barely wider than the radiating surface, the interference from waves diffracting off the outer edge is extremely low. The result is a smooth response and wide dispersion from the small radiating area.

Crossover. The active crossover frees the designer from the restrictions of limited drive unit sensitivity and response shape. So, it is a simple matter to provide any equalisation required and balance the system correctly. In addition, it allows the use of all-pass filters to alter the time delay of the drive units. Its active subtractive crossover network ensures flat crossover regions—amplitude and phase. The frequency response of a well-designed nautilus speaker system is shown in black line in Fig. 10. It should have no sudden large peak (shown in red) or sudden dips.

Driver units. The requirement for a nautilus shaped speaker system to be resonance-free demands that all the units should work like perfect pistons within their pass bands and up to two octaves above them as well. Perfect piston drive units are those that move without bending and with a total freedom from resonances. This means that the voice coil moves like a piston of an internal combustion engine, and so should the diaphragm with zero flex, that is, it is infinitely stiff.

To have perfect piston motion, a minimum of four drive units are needed to maintain as nearly a constant directivity as possible—because perfect pistons become increasingly directional at high frequencies, and because at a high enough frequency the diaphragms cease to work like a piston, displaying high Q resonances. These drivers, within their linear motion limits, add nothing to the sound and take nothing away, which means they simply reproduce exactly what is given to them, and so perfectly converting electrical energy into sound energy.

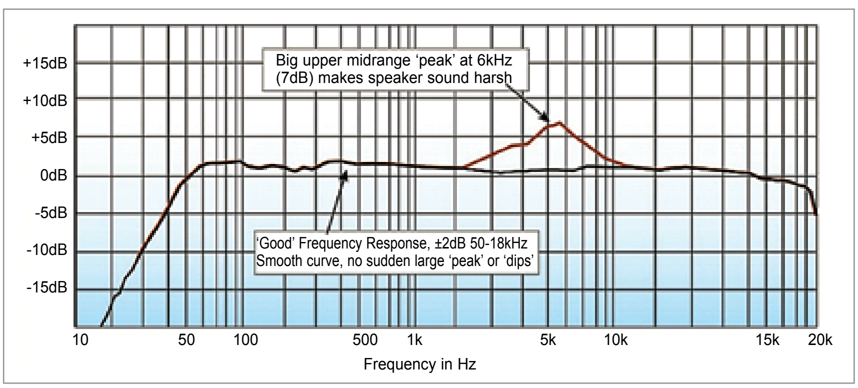

Fig. 10: Frequency response

Bass driver. The bass driver fires into a nautilus shaped coiled tube to save space. The unit's diaphragm is a perfect piston within its operating range. But it cannot be extended much higher in frequency due to resonance problem.

A single large pulp-coned driver for the bass frequency is very suitable. It has been found that aluminium cones larger than 200mm (8-inch) sound distinctly metallic and hence are not preferred. Therefore, a bass driver cone is made from a low-density pulp using Kevlar and stiffening resin. Kevlar is a high-strength, heat-resistant synthetic material, which is five times stronger than steel and is generally used for making bullet-proof vests.

The cone is obtained through slow deposition rather than high compression, followed by a surface impregnation of hard thermosetting resin that has internal structure of an open network of long undeformed fibres. This means that the overall structure is effectively a thick laminate with a very high resistance to the static bending stress and localised flexure. The vast improvement to bass performance is due to pistonic motion and use of improved driver cone. The cone has variable thickness, which is instrumental in maintaining linear pistonic behaviour.

The dust cap has a significant influence on sound quality. With a hard oversized dust cap domes offer the best performance with pure bass. However, with increase in size its resonance degrades the sound quality. This problem is minimised by using a carbon-fibre dust cone dome attached conventionally to the bass driver cone but driven directly by the elongated carbon fibre former. Sectional arrangement of the woofer is shown in Fig. 11.

Voice coil. The voice coil is made from high-quality enamelled copper wire (sometimes also made from silver wire). Only 5% of power delivered to the voice coil is converted to sound energy. The rest 95% of power is